

SHWETANK SINGH

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SUMMARY

Software Engineer at the intersection of backend systems and AI, tracing failures across microservices, turning unfamiliar codebases into LLM-assisted dependency maps, and building Java/Python services that hold up in production. The operating principle is operational truth over optimistic assumptions: instrument the system, validate the behavior, find the failure, and ship what works.

EDUCATION

California State University, Fullerton

Master of Science in Computer Science

Fullerton, CA

Aug 2024 – May 2026

Indian Institute of Information Technology

Bachelor of Technology in Computer Science and Engineering

Vadodara, India

Aug 2019 – Aug 2023

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, Solidity

Backend & Systems: RESTful APIs, FastAPI, Flask, Node.js, Express.js, Microservices, Distributed Systems

Databases: PostgreSQL, MySQL, MongoDB, Elasticsearch

Cloud & DevOps: GCP, AWS, Docker, Linux, CI/CD, Git

AI/ML: TensorFlow, CNNs, LLM Applications, AI-assisted Development

Observability: OpenTelemetry, Zipkin, Google Cloud Trace, Distributed Tracing, Logging, Monitoring, Postman

EXPERIENCE

Software Developer

Aug 2023 – Jul 2024

We Win Limited

Bhopal, India

- Built production backend services in **Java** and **Python** for business workflows and decision-support systems
- Developed and maintained **RESTful APIs** connecting application services, databases, and downstream components
- Designed ingestion, transformation, and validation pipelines to move application data reliably across backend services
- Optimized **SQL queries** and backend workflows across performance-sensitive operations
- Integrated monitoring and **distributed tracing**, cutting production issue resolution time by **30%**
- Resolved recurring production failures through log analysis, root-cause debugging, and stronger validation logic

Software Developer Intern

Jan 2023 – Jul 2023

We Win Limited

Bhopal, India

- Developed **REST APIs** for backend services, internal analytics tools, and external exchange integrations
- Built internal analytics dashboards for operational visibility into application data and workflows
- Integrated **blockchain exchanges** by mapping external API data into the platform and testing integration failures
- Worked with **SQL** and **MongoDB** to query, modify, and validate application data across services
- Instrumented microservices with **Zipkin**, **OpenTelemetry**, and **Google Cloud Trace** for request tracing

Software Developer Intern

May 2022 – Jul 2022

Voldemort With a Nose LLP

Hyderabad, India

- Developed an in-house **video editing application** using **Python** and **Tkinter** for arranging video shots
- Built an interactive editing interface and Python automation utilities to simplify repetitive media workflows
- Maintained internal software by debugging application issues and implementing functionality improvements

PROJECTS

CodeGraph AI — Interactive Repository Intelligence Platform

[GitHub](#)

- Built a repository explorer that turns source-code relationships into interactive dependency graphs using **React Flow**
- Developed a **Python FastAPI** backend for repository parsing, dependency analysis, and AI-generated architecture insights
- Combined structural analysis with **LLM explanations** to translate dependency graphs into plain-language architecture context
- Added dependency tracing and bottleneck detection to help developers navigate unfamiliar codebases
- Validated repository parsing and dependency analysis against Facebook's open-source React codebase

BCIRS — Decentralized Crime Incident Reporting Platform

- Built a hybrid architecture using **Solidity**, **IPFS**, and **MongoDB** for case state, evidence, and metadata
- Developed Citizen, Police, Judiciary, and Admin portals using **React**, **Node.js**, **Express**, and **Web3.js**
- Implemented role-based state transitions spanning incident reporting, investigation, judiciary review, and resolution
- Resolved ABI, network ID, routing, and state synchronization issues across blockchain and application layers

Bird Species Identification — CNN Image Classification System

- Built an end-to-end classifier using **Python**, **TensorFlow**, and CNNs across **200 species** and **11,788 images**
- Developed training and inference workflows achieving **93% training** and **80% testing accuracy**
- Built an image-upload workflow that processes inputs through the CNN and returns predicted species results

LEADERSHIP & ACTIVITIES

- **Teaching Associate, Compilers and Languages** — CSUF, Aug 2025 – Jan 2026: taught ~40 students in compiler design, debugging, and systems programming; recognized by faculty for exceptional scholastic ability and demonstrated academic achievement
- **Organizer, CSUF Google Developer Groups** — Aug 2025 – May 2026: led 3–4 sessions on backend systems, distributed architectures, and developer tooling
- **Dialership Program Participant, Dialpad** — San Ramon, CA, Aug 2025: discussed scalability, production systems, and software architecture with engineers and mentors